A Scalar-Collision Counterexample to Exact Continuous Unitary Convexity

Candidate full answer to Question 7.2 of arXiv:2210.13309

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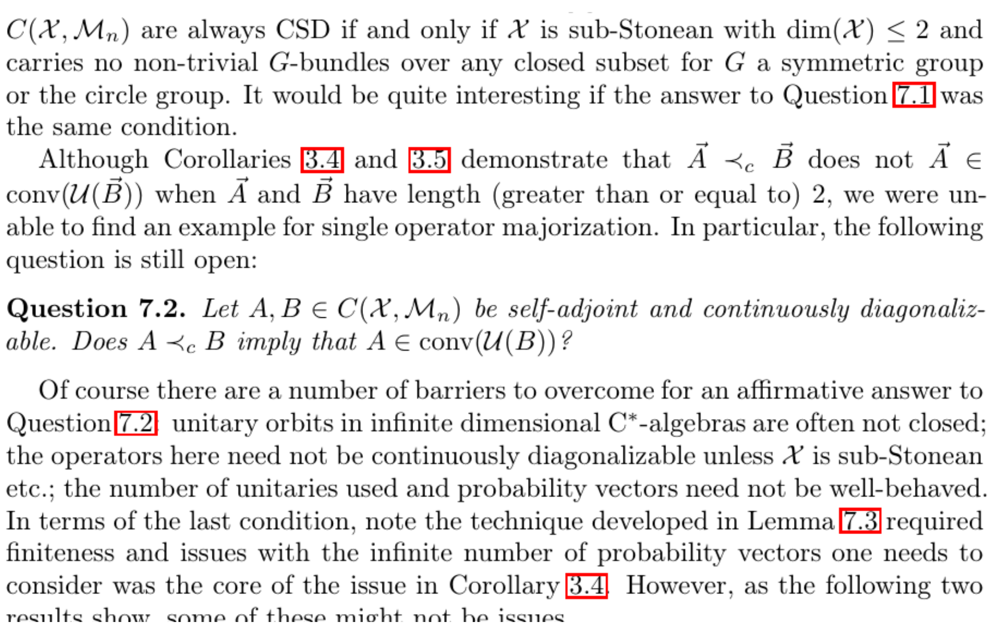
Status. Candidate full counterexample, likely valid, pending human review. There are diagonal self-adjoint fields $A, B \in C(X, M_3(\mathbb{C}))$ on a compact metric space such that $A \prec_c B$ but $A \notin \mathrm{conv}(\mathcal{U}(B))$. Thus Question 7.2 has a negative answer.

1 Source question

Mootoo and Skoufranis ask whether, for self-adjoint continuously diagonalizable fields $A, B \in C(X, M_n)$, continuous majorization $A \prec_c B$ always implies exact membership $A \in \mathrm{conv}(\mathcal{U}(B))$ [1, Question 7.2]. Exact membership means that there are finitely many continuous unitary fields and one scalar probability vector, independent of the base point.



$C(\mathcal{X}, \mathcal{M}_n)$ are always CSD if and only if $\mathcal{X}$ is sub-Stonean with $\dim(\mathcal{X}) \leq 2$ and carries no non-trivial G-bundles over any closed subset for G a symmetric group or the circle group. It would be quite interesting if the answer to Question 7.1 was the same condition.

Although Corollaries 3.4 and 3.5 demonstrate that $\vec{A} \prec_c \vec{B}$ does not $\vec{A} \in \mathrm{conv}(\mathcal{U}(\vec{B}))$ when $\vec{A}$ and $\vec{B}$ have length (greater than or equal to) 2, we were unable to find an example for single operator majorization. In particular, the following question is still open:

Question 7.2. *Let $A, B \in C(\mathcal{X}, \mathcal{M}_n)$ be self-adjoint and continuously diagonalizable. Does $A \prec_c B$ imply that $A \in \mathrm{conv}(\mathcal{U}(B))$?*

Of course there are a number of barriers to overcome for an affirmative answer to Question 7.2: unitary orbits in infinite dimensional C*-algebras are often not closed; the operators here need not be continuously diagonalizable unless $\mathcal{X}$ is sub-Stonean etc.; the number of unitaries used and probability vectors need not be well-behaved. In terms of the last condition, note the technique developed in Lemma 7.3 required finiteness and issues with the infinite number of probability vectors one needs to consider was the core of the issue in Corollary 3.4. However, as the following two results show, some of these might not be issues.

Figure 1: Question 7.2 in the published source, page 20.

The source gives a counterexample for a pair of operators by exploiting the edge between the identity and a three-cycle in the Birkhoff polytope, but leaves the single-operator question open. The construction below encodes the two operators as the two first-order directions of one operator at a scalar eigenvalue collision.

2 Construction and result

Let

$$P = \begin{bmatrix} 0 & 1 & 0 \\ 0 & 0 & 1 \\ 1 & 0 & 0 \end{bmatrix}, \qquad \alpha(s) = \frac{1+s}{3}, \qquad D_s = \alpha(s) I_3 + (1-\alpha(s))P$$

for $0 \le s \le 1$. Thus D_s is a nonconstant continuous path of doubly stochastic matrices and $1/3 \le \alpha(s) \le 2/3$.

Let $K = \{z \in \mathbb{R}^2 : \|z\| \le 1\}$, let $X = [0,1] \times K$, and write points of X as (s,z), with $z = (z_1, z_2)$. Define

$$b(z) = \begin{bmatrix} z_1 \\ z_2 \\ -z_1 - z_2 \end{bmatrix}, \qquad B(s,z) = \operatorname{diag}(b(z)), \qquad A(s,z) = \operatorname{diag}(D_s b(z)). \tag{1}$$

Theorem 1 (Negative answer to Question 7.2)**.** *The fields $A, B \in C(X, M_3(\mathbb{C}))$ in* (1) *are self-adjoint and continuously diagonalizable, and $A \prec_c B$. Nevertheless,*

$$A \notin \operatorname{conv}(\mathcal{U}(B)).$$

The exceptional set is the connected interval $[0,1] \times \{0\}$, on which $B = 0$. Away from that set, B has simple spectrum except along three lines in the z-disk. The obstruction is therefore concentrated at eigenvalue collisions, exactly outside the scope of the simple-spectrum affirmative result.

3 A support-rigidity lemma

A matrix $E = [e_{ij}] \in M_3(\mathbb{R})$ is *unistochastic* if $e_{ij} = |v_{ij}|^2$ for some unitary $V \in U(3)$.

Lemma 1 (The three-cycle edge is rigid)**.** *If E is unistochastic and*

$$\operatorname{supp}(E) \subseteq \operatorname{supp}(I_3 + P),$$

then $E = I_3$ or $E = P$.

Proof. Choose a unitary $V = [v_{ij}]$ with $e_{ij} = |v_{ij}|^2$. The support assumption gives

$$V = \begin{bmatrix} v_{11} & v_{12} & 0 \\ 0 & v_{22} & v_{23} \\ v_{31} & 0 & v_{33} \end{bmatrix}.$$

The first two rows overlap in only the second column, so their orthogonality gives $v_{12}\overline{v_{22}} = 0$. If $v_{12} = 0$, the row and column norm conditions successively force $|v_{11}| = |v_{33}| = |v_{22}| = 1$ and all other entries to vanish; hence $E = I_3$. If $v_{22} = 0$, they instead force $|v_{23}| = |v_{31}| = |v_{12}| = 1$ and all other entries to vanish; hence $E = P$. $\square$

4 Proof of Theorem 1

The fields in (1) are already diagonal, so they are self-adjoint and continuously diagonalizable. The map $(s,z) \mapsto D_s$ is a continuous doubly stochastic witness to

$$D_s b(z) = \operatorname{diag}(A(s,z)),$$

and therefore $A \prec_c B$ by the definition of continuous majorization.

Suppose, toward a contradiction, that $A \in \operatorname{conv}(\mathcal{U}(B))$. After deleting zero coefficients, there are $m \geq 1$, fixed numbers $t_j > 0$ with $\sum_j t_j = 1$, and continuous unitaries $U_j : X \to U(3)$ such that

$$A = \sum_{j=1}^{m} t_j U_j^* B U_j. \tag{2}$$

For $(s, z) \in X$, define

$$E_j(s, z)_{ik} = |U_j(s, z)_{ki}|^2 .$$

The transpose of a unitary is unitary, so every $E_j(s, z)$ is unistochastic; it is also doubly stochastic and varies continuously. Taking the diagonal of (2) gives

$$D_s b(z) = \sum_{j=1}^{m} t_j E_j(s, z) b(z). \tag{3}$$

Fix s and an arbitrary $z_0 \in \mathbb{R}^2$. For every sufficiently small $r > 0$, substitute $z = rz_0$ into (3), divide by r, and let $r \downarrow 0$. Continuity yields

$$D_s b(z_0) = \left(\sum_{j=1}^{m} t_j E_j(s, 0) \right) b(z_0). \tag{4}$$

The range of $b : \mathbb{R}^2 \to \mathbb{R}^3$ is the trace-zero plane

$$H = \{v \in \mathbb{R}^3 : v_1 + v_2 + v_3 = 0\}.$$

Thus (4) says that D_s and $F_s := \sum_j t_j E_j(s, 0)$ agree on H. Both matrices are doubly stochastic, so both send $\mathbf{1} = (1, 1, 1)^T$ to $\mathbf{1}$. Since $\mathbb{R}^3 = H \oplus \mathbb{R}\mathbf{1}$, we obtain the full matrix identity

$$D_s = \sum_{j=1}^{m} t_j E_j(s, 0). \tag{5}$$

Because $0 < \alpha(s) < 1$, the zero entries of D_s are exactly the three entries outside $\operatorname{supp}(I_3 + P)$. All t_j are positive and all entries of E_j are nonnegative, so (5) forces

$$\operatorname{supp}(E_j(s, 0)) \subseteq \operatorname{supp}(I_3 + P) \quad \text{for every } j, s.$$

Lemma 1 now gives $E_j(s, 0) \in \{I_3, P\}$. For fixed j, the map $s \mapsto E_j(s, 0)$ is continuous from the connected interval $[0, 1]$ into the discrete two-point set $\{I_3, P\}$; hence it is constant. The right side of (5) is therefore constant in s. The left side is not, because $\alpha(s) = (1 + s)/3$. This contradiction proves the theorem.

5 Mechanism and scope

The scalar collision itself erases all zeroth-order information: at $z = 0$ both A and B vanish. The two independent z-directions restore exactly the missing information. Passing to first order in those directions makes a hypothetical finite unitary decomposition recover the complete matrix D_s on the collision interval. Positivity then exposes its fixed zero pattern, and the unistochastic support lemma reduces every continuous summand to a discrete endpoint. Fixed scalar weights cannot reproduce a varying point on that edge.

This also explains why a continuous single Schur–Horn choice cannot repair the collision. More strongly, no finite ensemble with any fixed probability vector can repair it. The construction uses $n = 3$; the source proves the corresponding assertion affirmatively for $n = 2$.

6 Verification and literature boundary

The companion script `code/verify_construction.py` checks the doubly stochastic identities, the defining majorization equation, recovery of D_s from the two trace-zero directions plus $\mathbf{1}$, and the fact that the only permutation supports contained in $I_3 + P$ are I_3 and P. The support-rigidity step for general unistochastic matrices is the exact two-row orthogonality argument in Lemma 1, not a numerical inference.

Bounded exact-question, exact-title, keyword, and citation searches through 26 August 2026 found the source paper and results about *closed* convex hulls, but no later decision of Question 7.2. A recent paper on continuity of the Schur–Horn mapping proves local set-valued perturbation continuity [2]; it does not supply a global continuous finite unitary ensemble and does not address this exact-convex-hull obstruction. The theorem here should therefore be treated as a new candidate counterexample pending expert review.

7 Human review recommendation

The proof has two decisive checkpoints: verify that taking diagonals of (2) produces the transpose-modulus-square matrices E_j, and verify the limiting argument from (3) to (5). Once (5) is obtained, positivity, Lemma 1, and connectedness give the contradiction directly.

References

[1] X. Mootoo and P. Skoufranis, *Joint majorization in continuous matrix algebras*, J. Operator Theory **92** (2024), 413–438; arXiv:2210.13309.

[2] H. Chen and Y. Li, *On the continuity of Schur–Horn mapping*, arXiv:2407.00701, revised 4 January 2026.